Study & Defence Guide

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

Everything you need to understand your research, explain every number, present it in 12–15 minutes, and answer the committee's questions.

300	r = 0.47	88.7%	19%
participants, Baghdad	health anxiety ↔ cyberchondria	use AI chatbots	elevated health anxiety (SHAI ≥ 18)

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1. The study in 60 seconds

Say this if asked "summarise your research":
"Many people now look up their symptoms online, and increasingly they ask AI chatbots like ChatGPT before they see a doctor. We wanted to know how this relates to health anxiety and to cyberchondria, which is online searching that makes worry worse instead of better. We surveyed 300 adults in Baghdad using two validated scales, the SHAI-14 and the CSS-12. We found that AI chatbots are now a main source: 78.0% ticked them, 88.7% if we include those who named a chatbot. About 19% had elevated health anxiety. Health anxiety and cyberchondria were moderately correlated, r = 0.47, and health anxiety was the strongest predictor of cyberchondria, especially of excessive searching and distress after searching. The pattern is consistent with a cycle in which anxiety drives searching, searching drives distress, and distress drives extra doctor visits. Searching also shapes the consultation: about a third delay care because they feel reassured, and a third never tell their doctor about their searches. Because the study is cross-sectional, we cannot prove the direction of the cycle."

Aim

To study the relationship between health anxiety and cyberchondria and their correlates among adults in Baghdad who search for health information online, with attention to AI chatbots and pre-consultation behaviour.

Objectives (memorise the gist)

1. Describe online and AI health-information seeking and behaviour before consulting a doctor.
2. Estimate levels of health anxiety and cyberchondria and their association, independent of other factors (H1, H2).
3. Explore statistical pathways and modifiers (H3).
4. Link both to behaviour, disclosure and trust in doctors (H4).

H1–H2 are primary (confirmatory), H3–H4 secondary; everything else is exploratory.

Hypotheses and verdicts

Hypothesis	Verdict
H1: Health anxiety is positively associated with cyberchondria, most strongly with distress and excessiveness.	**Supported** (r = 0.47; excessiveness 0.46, distress 0.45)
H2: The association is independent of socio-demographic and digital factors.	**Supported** (partial r = 0.48; β = 0.48)
H3: Excessive searching mediates the link between health anxiety and post-search distress.	**Supported statistically, not causally** (indirect 0.16, 95% CI 0.11–0.23); reverse ordering fits equally well.
H4: Higher cyberchondria goes with care-seeking and a strained doctor relationship.	**Partly supported**: only higher cyberchondria → more disclosure survives FDR; urgent visits and strained relationship are exploratory.

2. Background concepts

Health anxiety

A continuum from normal worry to illness anxiety disorder (DSM-5). The cognitive-behavioural model (Salkovskis & Warwick, 1986) says people misinterpret normal body sensations as signs of serious disease; checking and reassurance-seeking reduce anxiety briefly but keep it going, because the person never learns the feared illness is not there. Lifetime prevalence of health anxiety disorder was 5.7% in an Australian national survey; almost half of Iraqi respondents reported health anxiety during COVID-19 restrictions (Karim et al., 2020).

Cyberchondria

Coined by White & Horvitz (2009): searches for common symptoms escalate towards rare, serious diagnoses. Starcevic & Berle (2013) define it as excessive or repeated online searching that *increases* distress. Searching itself is normal; it becomes cyberchondria when it is excessive, compulsive and distressing. It is seen as a safety behaviour maintained by intermittent reinforcement (sometimes the search reassures, so the habit is reinforced).

What was known before

- Meta-analyses: health anxiety correlates r ≈ 0.62–0.63 with cyberchondria and r ≈ 0.34 with general online searching (McMullan 2019; Schenkel 2021).
- One large longitudinal Dutch study (te Poel 2016) found a reciprocal loop in non-clinical health anxiety but not in clinical health anxiety.
- Almost no data from Iraq, and AI chatbots had not been studied alongside search engines.

The two instruments

Scale	Items / scoring	What higher means	Source
SHAI-14	14 items, each 4 statements scored 0–3; total 0–42. Cut-off ≥ 18 used for "elevated" health anxiety (a screening threshold, not a diagnosis; only 0.7% reach 29).	More health anxiety	Salkovskis et al., 2002; cut-off Österman et al., 2022
CSS-12	12 items, Never(1)–Always(5); total 12–60. Excessiveness (items 1–3), Compulsion (4–6), Distress (7–9), Reassurance (10–12).	More cyberchondria	McElroy et al., 2019; Arabic version Hallit et al., 2023

Know this detail: on the Google Form, some SHAI answer options were displayed in a different order from the original scale (e.g. "as much as others" appeared before "less than others"). We scored by the *meaning* of the chosen statement, following the original key, not by its position. Both our analysis and the independent analysis did this, and both produced identical scores.

3. Methods: what we did and why

Element	What we did	Why / how to justify
Design	Analytical cross-sectional online survey; STROBE and CHERRIES reporting.	Fast, affordable, appropriate for estimating associations; standard in this field.
Setting & time	Baghdad, 17 July 2026 – 11 September 2026.	Link shared in Baghdad-based online groups.
Sampling	Non-probability convenience + snowball; inclusion: ≥ 18 years, consent, Arabic or English.	No sampling frame of Internet users exists; the target behaviour (online searching) is by definition online.
Sample size	n = 300. Minimum 194 needed to detect r = 0.20 (α 0.05, power 80%).	A priori calculation only (post-hoc power is not meaningful); margin of error ±5.7% for proportions.
Instrument	Bilingual Google Form, 41 questions: consent, demographics, tools, motives, trust (1–5), SHAI-14, CSS-12, behaviours, disclosure, relationship, language.	Validated scales for the main constructs; bilingual to include English-preferring students.
Translation	Arabic wording by the team; equivalence checked empirically.	Arabic and English responders: same α, congruent loadings (Tucker φ ≈ 0.99), no mean difference.
Key definitions	Elevated health anxiety: SHAI ≥ 18 (not a diagnosis). High cyberchondria: CSS > 33 (top third). AI user: ticked AI *or* named a chatbot (32 named one without ticking).	SHAI cut-off from a diagnostic-accuracy study vs healthy controls; CSS-12 has no clinical cut-off.
Ethics	Voluntary, anonymous, e-consent required before any question; could withdraw any time.	[Insert approval committee and number.]

4. Statistics explained in plain language

For each analysis: **what it answers**, **why this test**, and **how to read the number**.

Analysis	What it answers	Why this test	How to read it
Cronbach's α, McDonald's ω	Do the items of each scale measure one thing consistently?	Standard reliability indices; ω is less restrictive than α.	SHAI α 0.81, CSS α 0.86: good (≥ 0.80).
Confirmatory factor analysis	Do CSS items group into the 4 subscales?	Tests the scale's structure in our population.	CFI 0.95, RMSEA 0.070: acceptable–good fit.
Shapiro–Wilk	Are scores normally distributed?	Decides parametric vs non-parametric tests.	$p < 0.05$, mild skew: so we used rank-based tests and robust SEs.
Pearson / Spearman correlation	How strongly do SHAI and CSS move together?	Both reported; Spearman robust to non-normality.	$r = 0.47$ = moderate; health anxiety "explains" 22% of cyberchondria variance.
Partial correlation	Does the link survive removing age, gender, background, chronic disease, trust, AI use?	Rules out simple confounding.	Partial $r = 0.48$: essentially unchanged.
Williams' t	Is SHAI more strongly linked to some CSS subscales than others?	Compares two correlations that share a variable.	Excessiveness and distress > compulsion and reassurance ($p < 0.001$).
Mann–Whitney U / Kruskal–Wallis H	Do scores differ between groups (gender, age, AI use…)?	Non-parametric group comparisons.	$p < 0.05$ = groups differ; effect size r_{rb} or ε^2.
χ^2 / Fisher exact	Is a category (e.g. social-media use) linked to clinical HA or high cyberchondria?	Categorical × categorical; Fisher when counts small.	Cramér's V ~ 0.1 small, 0.3 moderate.
Benjamini–Hochberg FDR	Which findings survive the many tests we ran?	162+ tests would produce false positives by chance.	$q < 0.05$ = robust; $q \geq 0.05$ = exploratory.
Hierarchical OLS regression (HC3)	What predicts cyberchondria when everything is considered together?	Adds blocks to see how much each explains; robust SEs.	R^2 0.030 → 0.133 → 0.317; SHAI $\beta = 0.48$ (largest).
Logistic regression	What predicts elevated health anxiety, or each behaviour?	Yes/no outcomes.	AOR > 1 = higher odds (e.g. CSS AOR 3.12 per SD).
Mediation (bootstrap)	Is HA linked to distress *through* excessive searching?	Tests a proposed pathway; 5000 resamples.	Indirect 0.16, CI excludes 0 = significant indirect path.
Moderation	Is the HA–CSS link stronger in some groups?	Interaction terms.	All $p > 0.05$: same link everywhere.
Sensitivity analyses	Are results fragile?	Remove flagged and careless responses, change cut-offs, bootstrap.	Main results unchanged.

If asked "why non-parametric tests?": "The Shapiro–Wilk test showed the scores were not perfectly normal, and several subscales were skewed, so we used rank-based tests for group comparisons. For

regression we used robust (HC3) standard errors and bootstrap intervals, and we also ran Welch's parametric tests as a check; the conclusions were the same."

If asked "why FDR correction?": "Because we tested every variable against every outcome, more than 160 tests. At $p < 0.05$ we would expect about 8 false positives by chance, so we controlled the false discovery rate and labelled anything that did not survive as exploratory."

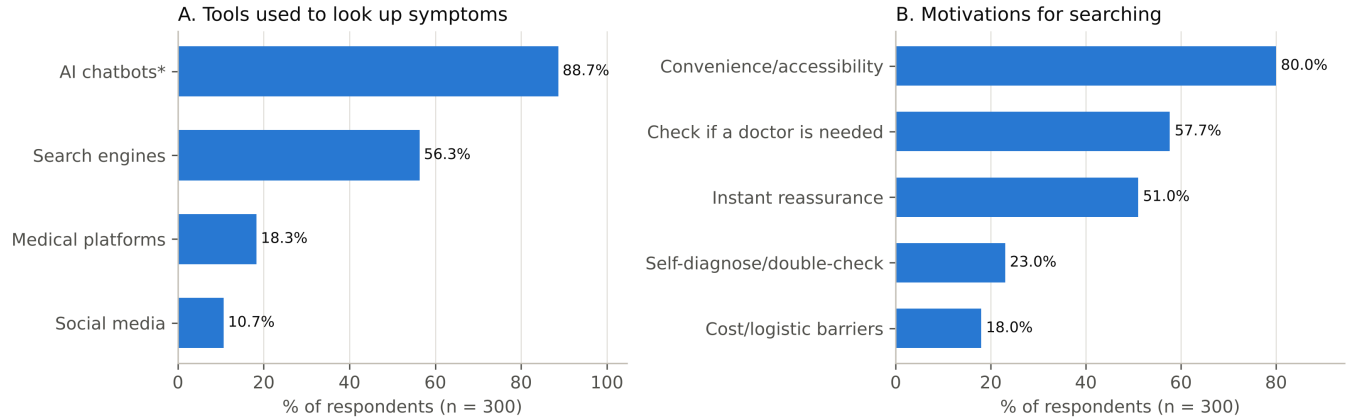
5. Results walkthrough

5.1 Sample

300 adults; 62.3% aged 20–29; 56.3% female; 44.0% medical students; 63.3% health-related background; 21.7% chronic condition; 70.7% answered mainly in Arabic.

> The sample is young and dominated by medical students: say so before the committee does.

5.2 How people search



*Includes 32 respondents who named an AI chatbot without ticking the AI option. Multiple answers allowed.

Figure 1 of the thesis: tools and motivations.

What to say: "AI chatbots have overtaken search engines: 78.0% ticked them (88.7% including those who named one) versus 56.3% for search engines, and 41.0% use AI without any search engine. ChatGPT dominates. People search mainly for convenience, to decide whether they need a doctor, and for instant reassurance."

5.3 What they do after searching

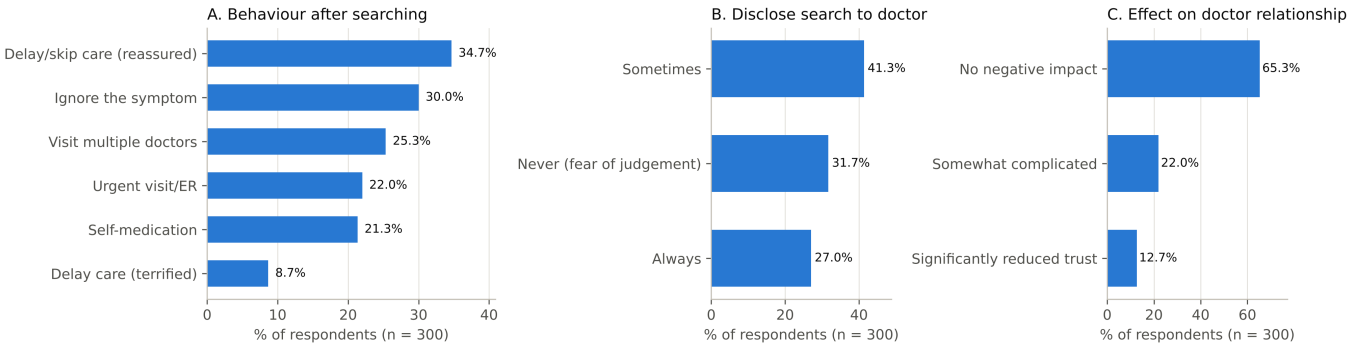


Figure 2 of the thesis: behaviour, disclosure and relationship.

What to say: "The most common response is to delay or skip care because the search reassured them (34.7%). Others go the opposite way: 22.0% seek urgent care and 25.3% see several doctors. Only 27.0% always tell their doctor what they found; 31.7% never do, citing fear of being judged. And 34.7% feel searching has complicated their relationship with doctors."

5.4 Levels of health anxiety and cyberchondria

SHAI mean 12.3 ± 5.8; 19% ≥ 18 (95% CI 15.0–23.8%). CSS mean 28.9 ± 8.4. Excessiveness highest, compulsion lowest: people search a lot, but it rarely disrupts their lives.

5.5 The main result: health anxiety and cyberchondria

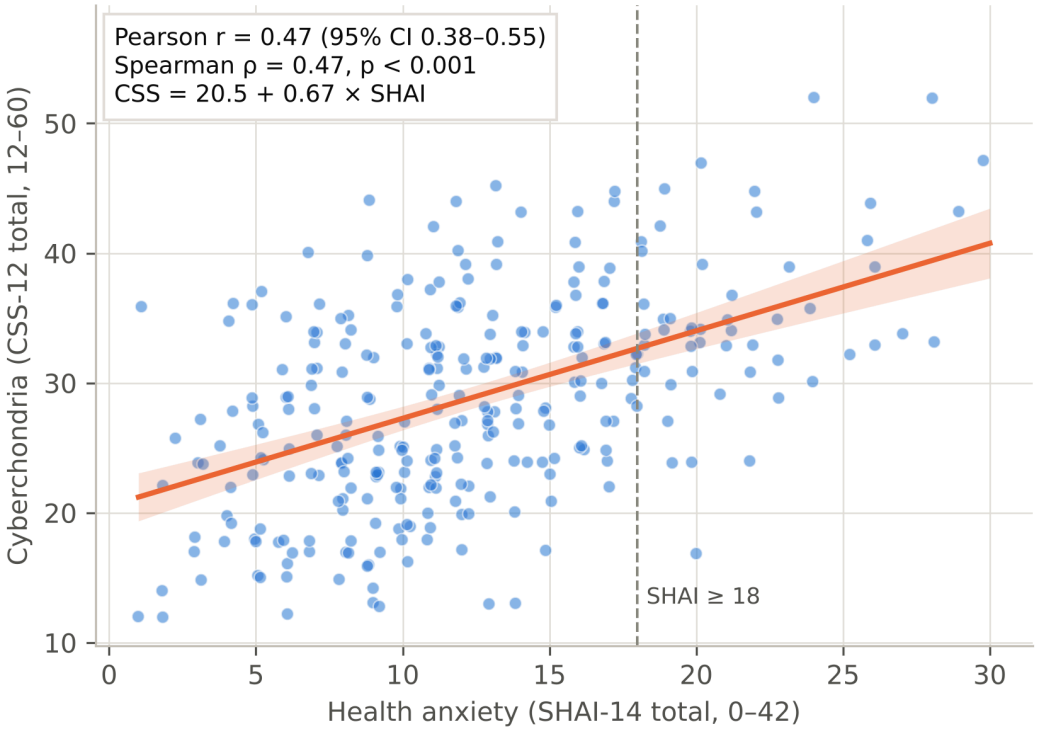


Figure 3 of the thesis: SHAI vs CSS.

What to say: "This is our main finding. Health anxiety and cyberchondria are moderately correlated, r = 0.47 (95% CI 0.38–0.55). The association hardly changes when we adjust for everything else (partial r = 0.48). It is strongest for excessive searching and distress after searching."

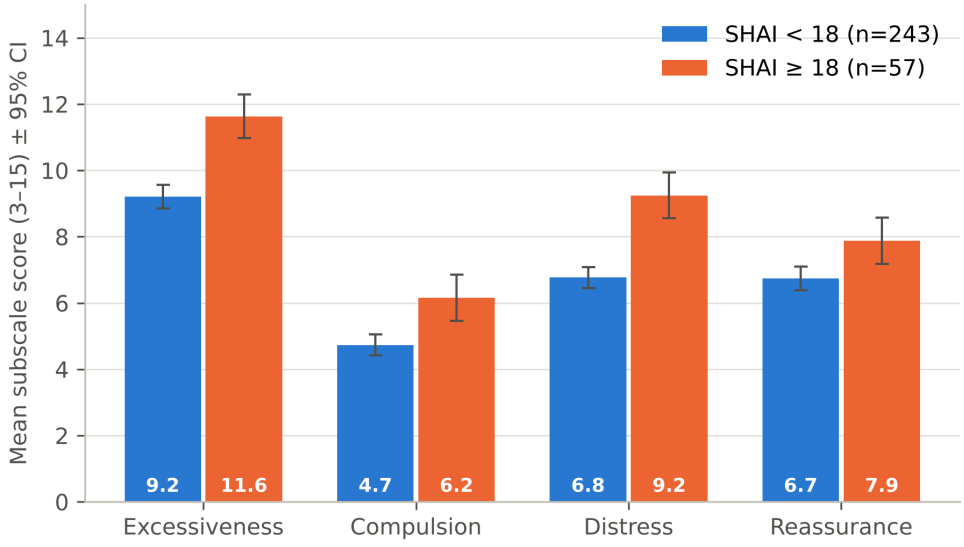


Figure 8 of the thesis: CSS subscales by health-anxiety status.

5.6 Which items drive the link

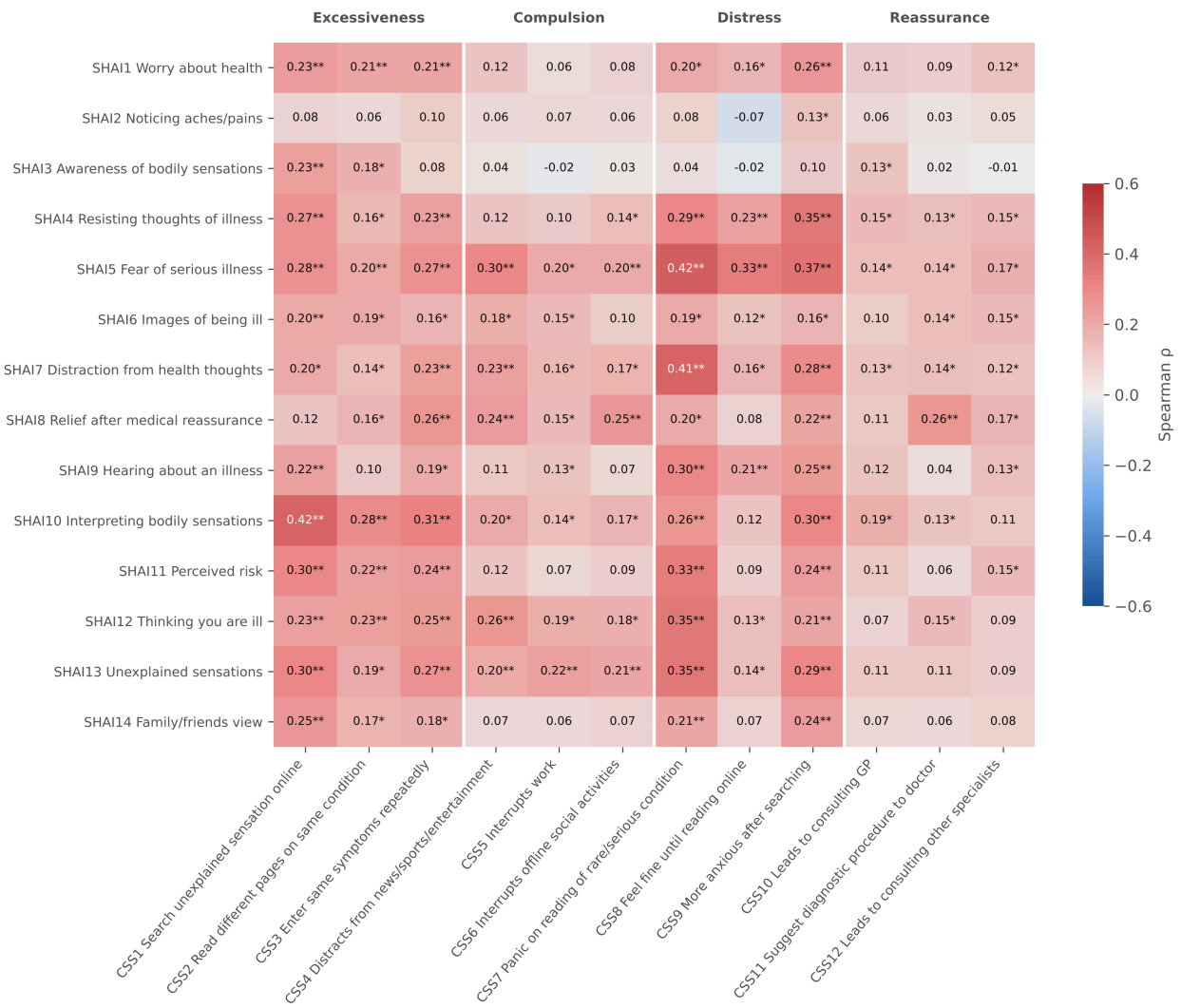


Figure 9 (appendix): item-by-item correlations.

What to say: "When we looked item by item, 111 of 168 pairs were significant after correction. The strongest links involve *fear* of serious illness and *interpreting* sensations, for example fear of illness with panic when reading about a rare disease. Simply noticing aches and sensations was not linked. So it is the catastrophic interpretation, not body awareness, that drives searching."

5.7 Who is affected (bivariate)

Health anxiety higher in: younger respondents, students, chronic illness, social-media searchers, and reassurance seekers. Gender: no difference. Cyberchondria: almost unrelated to demographics; higher in AI users and in those combining AI + search engines.

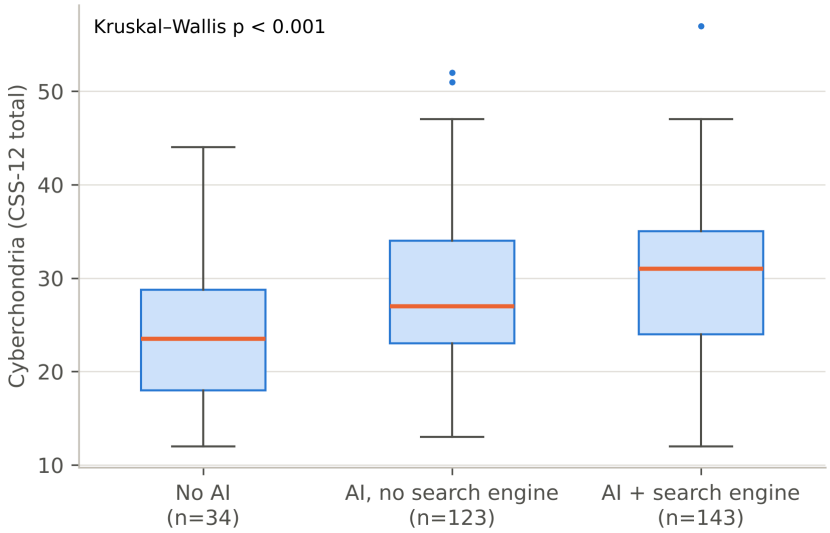


Figure 5 of the thesis: cyberchondria by tool profile.

5.8 Independent predictors

Cyberchondria model: demographics R^2 0.030; + digital use 0.133; + health anxiety 0.317 (ΔR^2 18.4%). SHAI B = 0.68 per point (β = 0.48). AI use B = +5.3 (95% CI 2.3–8.3).

Elevated health anxiety (logistic): cyberchondria AOR 3.12 per SD; reassurance motive AOR 3.83; each older age group AOR 0.61; AUC 0.83.

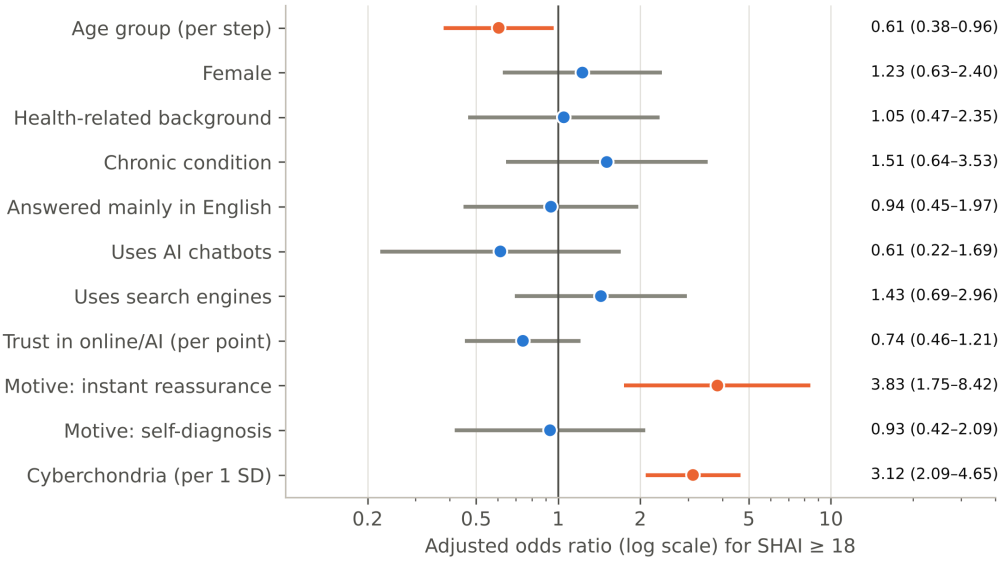
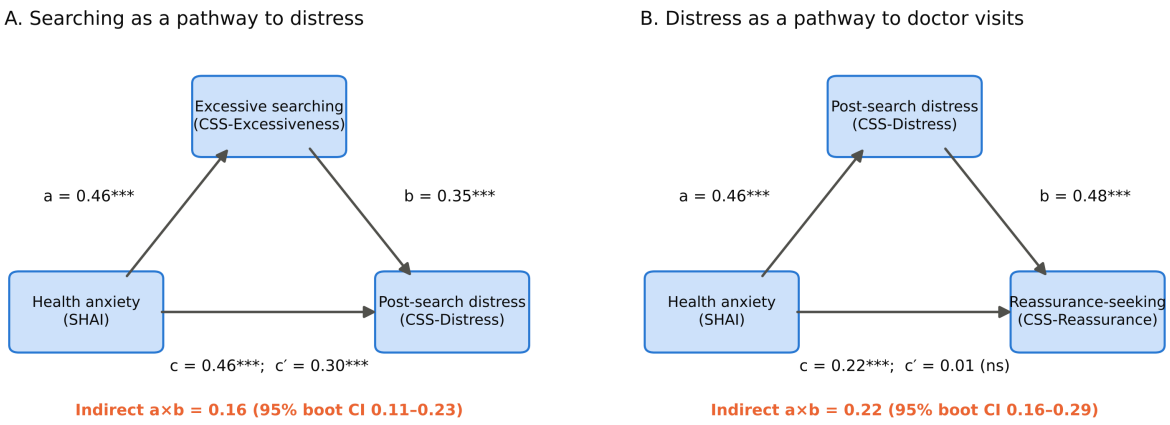


Figure 6 of the thesis: adjusted odds ratios for SHAI ≥ 18.

5.9 Pathways (mediation)



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

Figure 7 of the thesis: mediation models.

> **What to say:** "As an exploratory decomposition, most of the link between health anxiety and distress runs statistically through excessive searching (indirect 0.16), and distress statistically accounts for almost all of the link with reassurance-seeking visits (direct effect 0.01). But the reverse ordering fits just as well (0.15), and all the facets come from one questionnaire, so this is compatible with a cycle; it does not show one."

5.10 Behaviour models

Higher cyberchondria: more urgent visits (AOR 1.51), less ignoring (AOR 0.65). Disclosure: CSS AOR 0.57 for never disclosing (q = 0.012, survives FDR); SHAI AOR 1.47 (q = 0.098, exploratory). Higher trust: fewer multiple-doctor visits (AOR 0.59). Neither score predicts self-medication or reassured delay. Only the cyberchondria–disclosure result survives FDR; the rest are exploratory. Apparent AUC 0.83, optimism-corrected 0.79.

5.11 Moderation and robustness

No moderator changed the HA–CSS link (trust, AI use, background, gender, chronic disease, language, age). Medical students r = 0.57 vs others 0.40 (not significantly different). Excluding flagged or straight-line responders: r unchanged (0.46 / 0.47); dropping the reordered SHAI item 2 (13-item total, 0–39): r = 0.48.

6. Interpreting the findings

- **Why is our r (0.47) lower than meta-analyses (~0.62)?** Partly measurement error: corrected for reliability, r ≈ 0.56, close to the meta-analyses (but our result is still 0.47). Also a low-anxiety sample with a narrow range; young, digitally habitual users for whom searching is routine; most earlier studies used the 33-item CSS, which overlaps more with health anxiety.
- **Fear, not vigilance:** interpretation and fear items drive cyberchondria, which fits the cognitive model of health anxiety.
- **AI chatbots:** no reliable evidence of an association beyond health anxiety (the effect vanishes under the strict definition; 32 inconsistent answers; only 34 non-users). Say "absence of evidence", not "evidence of absence".
- **Two opposite behaviours:** searching pushes some people to complacency (delay, ignore) and others to over-use (urgent visits, multiple doctors). The reassured delay happens regardless of anxiety level, which is a safety concern.
- **Disclosure:** heavy searchers bring findings to the doctor (robust). Health-anxious people may hide searching for fear of being labelled "worried well", but that result is nominal only. The option itself mixed "never" with "fear of judgement". Clinicians should ask.
- **Medical students:** higher health anxiety only before adjustment (an age effect); identical cyberchondria to others, so medical knowledge does not protect.

7. Limitations and how to defend them

Limitation	How to answer
Cross-sectional: no causality	"Correct. We use the word association throughout, and we present mediation as consistent with, not proof of, a cycle. We tested the reverse direction too and recommend longitudinal work."
Convenience sample, 44% medical students, young	"Prevalence and AI-use figures cannot be generalised to all of Baghdad. But associations are less sensitive to sampling than prevalences, the HA–CSS link was the same in every subgroup, and medical students did not differ in cyberchondria."
Arabic not back-translated	"That was a limitation of our planning; validated Arabic versions exist and we recommend them. Internally, reliability was identical across languages, factor loadings were very similar (≈ 0.99, approximate structural similarity, not formal invariance) and scores did not differ, so we present results as conditional on equivalence."
SHAI cut-off from a Swedish study	"It is used as an indicator, not a diagnosis; it is a Swedish threshold separating patients from healthy controls. At the low-prevalence cut-off of 29 only 0.7% qualify, so clinically significant health anxiety is uncommon. We analysed SHAI as a continuous score."
Behaviour items not validated / multi-select	"They are descriptive, single items. We treat their associations as exploratory and say so."
Many tests	"We applied FDR correction and distinguished robust from exploratory findings."
AI effect depends on coding	"We report it transparently: with the strict definition the effect disappears, so it is a tentative finding."
Self-report, online	"Anonymous e-survey reduces social-desirability bias; careless responses were checked and excluding them changed nothing."

8. Numbers to memorise

Item	Value	Item	Value
Sample size	300	Data collection	Jul–Sep 2026
Aged 20–29	62.3%	Female	56.3%
Medical students	44.0%	Chronic condition	21.7%
Use AI chatbots	88.7%	Use search engines	56.3%
ChatGPT (of all)	59.3%	AI without search engine	41.0%
SHAI mean ± SD	12.3 ± 5.8	CSS mean ± SD	28.9 ± 8.4
SHAI ≥ 18	19%	α SHAI / CSS	0.81 / 0.86
SHAI–CSS r	**0.47**	Partial r	0.48
r with Excessiveness / Distress	0.46 / 0.45	r with Compulsion / Reassurance	0.25 / 0.24
SHAI β in CSS model	0.48	R^2 final model	0.317
Indirect effect (excess. → distress)	0.16	Direct HA → reassurance after distress	0.01 (ns)
Delay care (reassured)	34.7%	Never disclose to doctor	31.7%
Urgent care / ER	22.0%	Relationship complicated / reduced trust	34.7%
Item pairs significant (FDR)	111/168	References	52 (Vancouver)

9. Presentation plan: slide by slide (≈ 12–15 minutes)

Keep slides visual: one message per slide, the figure plus a one-line takeaway. Rehearse to time.

Slide 1. Title 15 s
Title, names, supervisor, college. Say the title clearly and introduce the team.

Slide 2. Why this matters 1 min
Everyone "Googles" symptoms; now AI chatbots answer them. For some people searching increases worry: cyberchondria. Health anxiety costs patients and services. No data from Iraq; AI not studied.

Slide 3. Key concepts 1 min
Define health anxiety and cyberchondria in one line each. Show the cycle: anxiety → search → distress → reassurance-seeking.

Slide 4. Aim and objectives 45 s
Aim + 3–4 condensed objectives; state H1–H4 briefly.

Slide 5. Methods 1.5 min
Cross-sectional online survey, Baghdad, n = 300; convenience/snowball; SHAI-14 and CSS-12 (show subscales); bilingual; sample-size justification (needed 194); ethics.

Slide 6. Analysis 45 s
One slide listing analyses: reliability/CFA, correlations, non-parametric tests with FDR, robust regression, logistic, mediation/moderation, sensitivity.

Slide 7. Participants 45 s
Key demographics; acknowledge 44.0% medical students.

Slide 8. How people search 1 min
Figure 1: AI 88.7% vs search engines 56.3%; motives.

Slide 9. Levels 45 s
SHAI mean 12.3; 19% ≥ 18; CSS mean 28.9; reliability α 0.81 / 0.86.

Slide 10. Main result 1.5 min
Scatter plot: r = 0.47, partial r = 0.48; subscale correlations; regression β = 0.48.

Slide 11. What drives it 1 min
Item-by-item heatmap: fear and interpretation, not awareness.

Slide 12. Pathways 1.5 min
Mediation figure; say "exploratory decomposition, compatible with a cycle, not proof".

Slide 13. Behaviour and the doctor 1.5 min
Delay 34.7%, never disclose 31.7%, urgent care; disclosure (cyberchondria → more disclosure is the robust result).

Slide 14. AI chatbots 45 s

Both prevalences (78% ticked, 89% incl. named); association not robust to classification; no reliable evidence beyond health anxiety.

Slide 15. Limitations 1 min

Cross-sectional, sample, translation, single items; how each was mitigated.

Slide 16. Conclusions and recommendations 1 min

Three conclusions + recommendations for clinicians, public health, AI developers, research.

Slide 17. Thank you / questions -

Keep the main figure on screen while taking questions.

10. Anticipated committee questions with model answers

About the design and sample

Q. Why a cross-sectional design?

It is the most feasible design for a graduation project and suitable for measuring associations. We are clear that it cannot establish causality; a longitudinal design is recommended.

Q. How do you know participants were from Baghdad?

We do not; residence was not verified. Recruitment channels were Baghdad-based, which we state as a limitation.

Q. Is your sample really patients?

No. It is a convenience sample of mostly young adults with a health background, which is why we revised the title.

Q. Your sample is mostly medical students. Is that a problem?

It limits generalisability of percentages such as AI use and prevalence. But the main association was the same across all occupational groups, medical students did not differ in cyberchondria, and we adjusted for occupation in all models.

Q. How did you calculate the sample size?

For the primary objective, a correlation: detecting r = 0.20 with 80% power and α = 0.05 requires 194. We exceeded that with 300. We did not report post-hoc power, which reviewers discourage. For proportions, the margin of error is ±5.7%.

Q. Why did you include participants who answered in English?

Many students in Baghdad study in English. Offering both languages improved accessibility. We checked that both language groups responded equivalently.

About the instruments

Q. Why the SHAI-14 and the CSS-12?

They are the most widely used validated measures of each construct, they are short (good for online surveys), the CSS-12 was validated against the SHAI, and both have Arabic versions in the literature.

Q. Why did you not use the validated Arabic CSS-12 and SHAI?

That was a limitation of our planning. We produced our own wording and tested it internally: α 0.81 and 0.86, equal reliability across languages and very similar factor loadings (Tucker φ ≈ 0.99). That is not formal invariance, so we present the results as conditional, and we recommend the Hallit 2023, Alshaibani 2023 and Alshayea 2020 versions for future work.

Q. Why a cut-off of 18?

It was the optimal SHAI-14 cut-off for distinguishing health anxiety from healthy controls in a 2022 diagnostic-accuracy study. It is an indicator, not a diagnosis, so we also report 15 and 22 and use the continuous score.

Q. What about the answer-order difference on the form?

Only one SHAI item (item 2, noticing aches and pains) was shown out of order. We scored it by meaning, and dropping it entirely left the result unchanged (r = 0.48 with the 13-item total).

Q. Why does CSS item 3 mention only search engines?

It is the original wording. We acknowledge it may under-capture repeated chatbot querying and recommend adapting it to "search engine or AI chatbot".

About the analysis

Q. Why both Pearson and Spearman?

Pearson is standard and comparable with meta-analyses; Spearman is robust to the mild non-normality. Both gave 0.47.

Q. What does r = 0.47 mean in practice?

A moderate positive association: people with more health anxiety tend to have more cyberchondria. Health anxiety accounts for about 22% of the differences in cyberchondria.

Q. What is FDR and why use it?

False discovery rate correction controls the proportion of false positives when running many tests. We tested every variable, so without it some significant results would be chance findings.

Q. Does your mediation show a mechanism?

No. It is an exploratory decomposition using subscales of one questionnaire at one time point. Excessive searching statistically accounts for part of the link with distress (indirect 0.16), but the reverse model fitted similarly, so following Maxwell & Cole and Fiedler et al. we treat it as exploratory.

Q. Can mediation in cross-sectional data prove a cycle?

No. It tests whether the data are consistent with a proposed ordering. We also tested the reverse ordering, which was also significant, which is consistent with a two-way relationship but cannot prove direction.

Q. Why robust (HC3) standard errors?

The residuals showed some heteroskedasticity, and robust standard errors remain valid in that situation.

Q. Did you check multicollinearity?

Yes: the highest VIF was 3.9, below the usual threshold of 5–10.

Q. How robust are the results?

Excluding the three annotated responses, excluding nine straight-line responders, bootstrapping, and changing the cut-off did not change the main results. An independent analysis of the same data reached the same conclusions.

About the findings

Q. Does AI use cause cyberchondria?

Our data cannot show causation, and the association was not robust: it disappeared (B = 1.5, p = 0.31) when only respondents who ticked AI were counted. We conclude there is no reliable independent association beyond health anxiety.

Q. Why reclassify the 32?

They named a specific chatbot, which is direct evidence of use. Because the decision changes the result, we report both definitions.

Q. Why is your correlation lower than in previous studies?

Ours is uncorrected and from a non-clinical, range-restricted sample, and we used the CSS-12 rather than the 33-item CSS. Corrected for the reliability of both scales it would be about 0.56, close to the meta-analytic 0.62–0.63; our actual result remains 0.47.

Q. CFI = 1.000: is that not too good?

That was the DWLS estimate, whose indices are inflated relative to ML cut-offs. We base our conclusions on ML (CFI 0.954, RMSEA 0.070, SRMR 0.053).

Q. Is the logistic model overfitted?

With about five events per predictor the estimates are optimistic. The apparent AUC was 0.83; corrected for optimism by bootstrap it was 0.79, and we treat the model as exploratory.

Q. Why tertiles for "high cyberchondria"?

No validated CSS-12 cut-off exists. The tertile is a relative category; our primary analyses use the continuous score.

Q. What is the most clinically important finding?

That a third of people delay care because they feel reassured, and a third say they never tell their doctor about their searches for fear of judgement. Doctors should ask about online and AI searches openly and without judgement.

Q. What does "never disclose" mean if the option includes fear of judgement?

It captures non-disclosure attributed to fear of judgement, so we interpret it that way. Higher cyberchondria robustly predicted more disclosure ($q = 0.012$); the link between health anxiety and hiding was nominal only ($q = 0.098$).

Q. Do medical students have more health anxiety?

Only before adjustment; the difference disappeared after accounting for age. Their cyberchondria was identical to others, so medical knowledge did not protect against it.

Q. What would you do differently?

Use random sampling, add a residence question, formally back-translate and pilot the Arabic version, measure behaviour frequency rather than tick boxes, and follow participants over time.

Q. What is new in your study?

To our knowledge, the first study to apply the CSS-12 in a Baghdad-based sample; one of the first studies to place AI chatbots alongside search engines; linking both to concrete pre-consultation behaviours and disclosure to doctors.

Q. What are your recommendations?

Clinicians: ask about searches non-judgementally and recognise repeated searching with distress as a sign of health anxiety. Public health: Arabic digital/AI health-literacy campaigns. AI developers: anxiety-aware responses. Researchers: longitudinal studies and formal Arabic validation.

11. Glossary

Term	Meaning
Health anxiety	Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by checking and reassurance-seeking.
Cyberchondria	Excessive or repeated online health searching that increases (rather than relieves) anxiety or distress.
SHAI-14	Short Health Anxiety Inventory, 14 items each scored 0–3 (total 0–42). Higher = more health anxiety.
CSS-12	Cyberchondria Severity Scale short form: 12 items scored 1–5 (total 12–60), four 3-item subscales.
Excessiveness	CSS subscale: searching a lot (search every sensation, read many pages, repeat the same search).
Compulsion	CSS subscale: searching interferes with work, leisure and social life.
Distress	CSS subscale: panic and anxiety caused by what is read online.
Reassurance seeking	CSS subscale: searching leads to visits to GPs/specialists or requests for tests.
Cross-sectional study	All data collected at one point in time; shows associations, not cause and effect.
Convenience / snowball sampling	Non-random recruitment of whoever is reachable, with participants sharing the link onwards.
Pearson r / Spearman ρ	Correlation coefficients from −1 to +1. Spearman uses ranks and does not assume normality. 0.1 small, 0.3 moderate, 0.5 large.
Partial correlation	The correlation that remains after statistically removing the effect of other variables (covariates).
Mann–Whitney U / Kruskal–Wallis H	Non-parametric tests comparing two / three-or-more groups on a score.
Chi-square (χ^2) / Fisher exact	Tests whether two categorical variables are associated; Fisher is used when counts are small.
Effect size	How big a difference or association is (e.g. r, ε^2, Cramér's V, β, OR), independent of sample size.
p-value	Probability of a result at least this extreme if there were truly no association. $p < 0.05$ = "statistically significant".
FDR / q-value	False discovery rate correction (Benjamini–Hochberg) for running many tests; $q < 0.05$ means the finding survives the correction.
Holm–Bonferroni	A stricter multiple-testing correction controlling the chance of any false positive.
Linear (OLS) regression, B and β	Predicts a score from several variables at once. B = change in outcome per unit of predictor; β = the same in standard-deviation units (comparable across predictors).
R^2 / ΔR^2	Proportion of variance in the outcome explained by the model / by an added block of variables.
HC3 robust standard errors	Standard errors that stay valid when the spread of residuals is uneven (heteroskedasticity).
Logistic regression, AOR	Predicts a yes/no outcome. Adjusted odds ratio > 1 = higher odds, < 1 = lower odds, after accounting for other variables.
AUC	How well a model separates people with and without the outcome: 0.5 = chance, 0.8+ = good.

Mediation	Tests whether X is linked to Y through a middle variable M (X → M → Y). Indirect effect = a × b.
Bootstrap	Re-sampling the data thousands of times to get confidence intervals without strong assumptions.
Moderation / interaction	Tests whether the strength of an association differs by a third variable (e.g. by gender).
Cronbach's α / McDonald's ω	Internal consistency (reliability) of a scale; ≥ 0.70 acceptable, ≥ 0.80 good.
CFA (CFI, TLI, RMSEA)	Confirmatory factor analysis checks whether items group as the scale claims. Good fit: CFI/TLI ≥ 0.95 (≥ 0.90 acceptable), RMSEA ≤ 0.06–0.08.
95% CI	Range of values compatible with the data; if it excludes 0 (for B, r) or 1 (for OR), the result is significant.

12. Final checklist before the defence

- ☐ Fill in the title page, acknowledgment and ethics approval details in the thesis.
- ☐ Read the full thesis once and the Results and Discussion twice.
- ☐ Memorise the "60-second summary" and the numbers table (section 8).
- ☐ Prepare slides following section 9; one message per slide; rehearse to 12–15 minutes.
- ☐ Practise answering every question in section 10 aloud, ideally with a colleague playing the committee.
- ☐ Be ready to explain: cross-sectional ≠ causal; what r, p, q, AOR and indirect effect mean.
- ☐ Bring a printed copy of the one-page summary and the questionnaire.
- ☐ If you don't know an answer: say so honestly, then say how you would find out. Committees respect that.